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Expression of Fibroblast Growth Factor-2 in the Nucleus Ambiguus Following Recurrent Laryngeal Nerve Injury in the Rat

Tetsuji Sanuki, MD; Eiji Yumoto, MD; Masahiro Komori, MD; Masamitsu Hyodo, MD

Objectives: **To examine fibroblast growth factor-2 (FGF-2) immunoreactivity in the nucleus ambiguus (NA) after three different recurrent laryngeal nerve (RLN) injuries.** ***Study Design:*** **Immunohistochemical analysis of FGF-2.** ***Methods:*** **Thirty adult rats underwent left-sided RLN crush (group A). The left RLN was transected in groups B (n = 30) and C (n = 30); in group C, both nerve stumps were covered with silicone caps. FGF-2 in the NA was assessed as the ratio of the positive areas on the left (operated [*O*]) and right (unoperated [*U*]) sides. The ratio (*O/U*) was measured 1, 3, 7, 14, and 28 days after the procedure. Three rats underwent left-sided RLN exposure and were killed 7 days later (control).** ***Results:*** **Left-sided RLN paralysis occurred until day 28 in group A. In the control group, *O/U* was approximately 1. In group A, *O/U* was significantly elevated on day 7; in group B, on days 3, 7, and 14; and in group C, on day 3. *O/U* in group B was significantly greater than that in group A on days 14 and 28. Maximal FGF-2 immunoreactivity was significantly lower in group C than in groups A and B.** ***Conclusions:*** **We demonstrated elevated production of FGF-2 in the NA after RLN injury. This endogenous FGF-2 might contribute to preventing lesion-induced neuronal death. Blockage of axonal regeneration might suppress FGF-2 production in the NA. Further understanding of the roles of FGF-2 after RLN injury may contribute to the prevention of neuronal death and facilitation of axonal regeneration.** ***Key Words:*** **Recurrent laryngeal nerve, fibroblast growth factor-2, nucleus ambiguus, nerve injury, immunohistochemistry.**

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INTRODUCTION

Unilateral recurrent laryngeal nerve (RLN) paralysis, which causes breathy dysphonia or mild aspiration, is an important clinical entity in otolaryngology. Its etiology includes varying degrees of injury to the nerve resulting from trauma and viral infection[1] or the direct invasion of malignant tumors arising from various organs. Steroids and vitamin B complex are administered orally to patients without a history of nerve transection. Although reinnervation of the laryngeal muscles occurs in some patients several weeks or months after the onset, other patients do not recover. Therefore, the factors that prevent neuronal death and facilitate nerve regeneration are of great clinical interest.

Axonal injury results in wallerian degeneration of the peripheral nerve, followed by neuronal death.[2–4] Axonal regeneration after axotomy requires axonal sprouting initially and growth of the axons thereafter. Several proteins promote neuron survival and growth of the regenerated axon. These include fibroblast growth factor-2 (FGF-2), ciliary neurotrophic factor (CNTF), nerve growth factor (NGF), brain-derived neurotrophic factor (BDNF), and insulin-like growth factor (IGF). These proteins are thought to be produced by the neuron itself, glial cells in the central nervous system, or Schwann cells in the peripheral nerve and target muscle. In rodents, after peripheral axotomy some motoneurons survive, whereas others proceed to cell death.[5,6] Facial motoneurons of immature rats are particularly sensitive to axotomy, with more than 80% loss of neurons occurring within 1 week after facial nerve injury in newborn rats.[7] Neuronal death after axotomy occurs within 12 hours of sciatic nerve crush in immature rats.[8] Most cases of neuronal death after sciatic nerve transection in neonatal rats were prevented by intraperitoneal administration of FGF-2.[9] In adult rats, administering FGF-2 to the site of nerve injury or the nucleus prevented neuronal death after nerve transection.[10,11] In 1999, Chen et al.[12] showed that local treatment with the FGF-2-neutralizing antibody or an antagonist for the FGF-2 receptor caused a decrease in the number of regenerated axons and suppressed the growth of regenerating axons following facial nerve transection, indicating a neurotrophic role of endogenous FGF-2. However, FGF-2 expression in the nucleus following varying degrees of nerve injury has not been elucidated.

From the Department of Otolaryngology (T.S., M.K., M.H.), Ehime University School of Medicine, Japan; and the Department of Otolaryngology—Head and Neck Surgery (E.Y.), Kumamoto University School of Medicine, Kumamoto, Japan.

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Send Correspondence to Tetsuji Sanuki, MD, Department of Otolaryngology, Ehime University School of Medicine, Shigenobu-cho, Onsen-gun, Ehime, 791-0295, Japan.

Because the RLN is relatively thin and thus difficult to manage, Komori's 1999 report[13] is the only one on the effects of FGF-2 on regeneration of the RLN after injury. Komori reported that local administration of FGF-2 significantly facilitated functional and morphological recovery of the injured RLN in rats compared with other neurotrophic factors, such as CNTF and NGF. Our ultimate goal is to examine the roles of FGF-2 in preventing neuronal death in the nucleus ambiguus (NA) and facilitating axonal regeneration after varying degrees of RLN injury. The purpose of this study was to determine FGF-2 expression in the NA after three different nerve injuries in adult rats: nerve crush using needle forceps, which corresponded to a Sunderland grade II injury,[14] and transection of the nerve with or without placing a cap on the stump of the nerve to prevent regeneration. FGF-2 in the NA was detected immunohistochemically and assessed quantitatively 1, 3, 7, 14, and 28 days after injury.

MATERIALS AND METHODS

Ninety-three 10-week-old male Wistar rats were used in this study. All the animals were housed at a constant temperature (22°C) with a 12-hour:12-hour (12:12) light–dark cycle and given food and water ad libitum. The following experiments were conducted in accordance with the guidelines of the Animal Care and Use Committee in compliance with the Animal Research Center of Ehime University School of Medicine, Ehime, Japan.

Recurrent Laryngeal Nerve Injury

Surgical procedures were performed under aseptic conditions while the animals were anesthetized with intraperitoneal sodium pentobarbital (30 mg/kg body weight). Following injection of 1% lidocaine hydrochloride into the anterior aspect of the neck, a vertical skin incision was made in the neck and the left-sided RLN was carefully exposed under an operating microscope (Olympus OME, Tokyo, Japan). The animals were divided into four groups. In group A (n = 30), 1 mm of the left-sided RLN at the level of the seventh tracheal ring was crushed with needle forceps by 2.0 kg/cm^2 pressure for 5 seconds (Fig. 1A). In group B (n = 30), 5 mm of the left-sided RLN was resected at the level of the seventh tracheal ring. The central and peripheral nerve stumps received no additional treatment, and the injured nerve was repositioned so there was 5 mm between the two cut ends (Fig. 1B). In group C (n = 30), the left-sided RLN was resected similarly, but both nerve stumps were covered with sterile silicone caps (1.2 mm, inner diameter [ID]; 2.0 mm, outer diameter [OD]; length, 5.0 mm) (Dow Corning Co., Midland, MI). The caps were fixed to the perineurium of the nerve using two 8-0 nylon sutures (Fig. 1C). In group D (n = 3), the left-sided RLN was exposed and there was no further treatment. Animals in this group served as controls. Following completion of these procedures, the skin incision was closed and the rats were allowed to recover in an approved animal care facility.

On days 1, 3, 7, 14, and 28 after the procedure, six animals each in groups A, B, and C were killed for the immunohistochemical study as described below.

Videoendoscopic Evaluation of Vocal Fold Movement

An anesthetized rat was secured on an operating platform in the supine position. A rigid endoscope (SFD, Nagashima Medical Instruments Co., Tokyo, Japan) was inserted orally and adjusted to provide the best view of the larynx. The endoscope was connected to a charge-coupled device (CCD) camera (WV-KS152, Panasonic, Osaka, Japan). Before RLN injury and on the day the animals were killed, vocal fold movement in each animal was recorded with a video cassette recorder (AU-W35R, Panasonic) to evaluate vocal fold movement.

Immunohistochemical Staining

On days 1, 3, 7, 14, and 28 after the procedure, the animals were anesthetized in the same manner as described above and then perfused transcardially with 4% paraformaldehyde in 0.1 mol/L phosphate buffer (pH 7.4). After confirming cardiac arrest, the brainstem was immediately harvested from each rat and postfixed for 5 minutes in the same fixative. The specimens were transferred to 0.1 mol/L phosphate buffer (pH 7.4) containing 20% sucrose overnight at room temperature. The medulla oblongata including the NA, which innervates the RLN, was serially

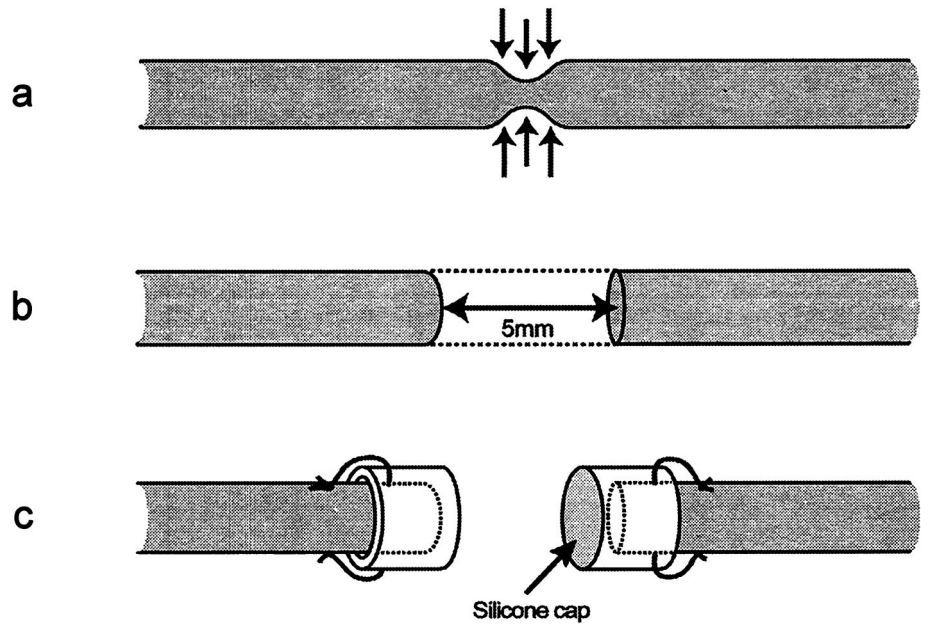


Fig. 1. Procedures used to produce recurrent laryngeal nerve injury. **(A)** Group A: 1 mm of the left-side recurrent laryngeal nerve (RLN) was crushed with needle forceps for 5 seconds at the level of the seventh tracheal ring. **(B)** Group B: 5 mm of the left-sided RLN was resected at the level of the seventh tracheal ring. **(C)** Group C: 5 mm of the left-sided RLN was resected at the level of the seventh tracheal ring, and both nerve stumps were covered with sterile silicone caps.

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cut in 20-μm-thick sections in the transverse plane in a cryostat at −20°C. The sections were incubated with 0.1 mol/L phosphate-buffered saline (pH 7.6) containing 0.25% carrageenan, 5% bovine serum albumin (BSA), 1% normal goat serum, and 0.1% Triton X-100 for 1 hour at room temperature. They were further incubated with anti-FGF-2 mouse antibody (Transduction Laboratories, Lexington, KY) for 45 minutes. After washing in 0.01 mol/L phosphate-buffered saline (pH 7.4) containing 0.2% Triton X-100, the sections were processed for biotinylated rabbit anti-mouse immunoglobulin G-1 (IgG-1) antibody (DAKO AIS, Glostrup, Denmark) for 30 minutes and further processed with avidin-biotin-horseradish peroxidase complex, (DAKO Co., Carpinteria, CA) for 10 minutes. The sections were stained with a solution of 3,3-diaminobenzidine (0.014%) and H_2O_2 (0.007%) in 0.01 mol/L phosphate-buffered saline (pH 7.4) for 30 minutes, dehydrated through a graded ethanol series, and mounted in Canada balsam (Merck, Darmstadt, Germany).

Quantification of Fibroblast Growth Factor-2 Immunoreactivity by Computer Image Analysis

Computer-assisted imaging analysis was used to determine the relative FGF-2 immunoreactivity in the NA by comparing the left side with the unoperated right side. Each pair of sections was observed under a microscope (Optiphot, Nikon Inc., Tokyo, Japan) equipped with a Digital still camera (HC-300, Fujifilm, Tokyo, Japan). Microscopic images of the medulla oblongata including the NA were imported into a personal computer (Power Macintosh 7600, Apple, Cupertino, CA) and stored using image-capturing software (Photograb-300, Fujifilm). The captured images were cropped to the size of the NA in an image of at least 100 $\times$100 pixels. The public-domain software NIH Image was used to quantitatively determine the expression of FGF-2 immunoreactivity. The numbers of pixels that were positive for FGF-2 immunoreactivity on the operated (*O*) and unoperated (*U*) sides were counted automatically. The quantity of FGF-2 expression in the NA induced by RLN injury was represented as the ratio of the positive area on the O side and the U side (O/U).

Statistical Analysis

The data were represented as means ± standard deviation. Paired and unpaired Student *t* tests were used to compare the FGF-2 immunoreactivity between the *O* and *U* sides in the same group and among three groups. The statistical analysis was performed using statistical software (StatView, Abacus Concepts, Inc., Berkeley, CA). A *P* value less than .05 was taken as the level of significance.

RESULTS

Vocal Fold Movement

In the videoendoscopic study, left-sided vocal fold paralysis was observed on days 1, 3, 7, 14, and 28 after the procedure in groups B and C. In group A, the left-sided vocal fold was fixed on days 1, 3, 7, and 14 after the procedure, but vocal fold movement recovered by day 28 after the nerve crush. Left-sided vocal fold paralysis was not found in group D.

Immunohistochemical Staining of Fibroblast Growth Factor-2 in the Nucleus Ambiguus

All animals in group D (control group) were killed on day 7 after the procedure. FGF-2 immunoreactivity was seen as a brown color in the neuronal perikarya and nerve fibers in the NA. In addition, FGF-2 immunoreactivity was found in relatively small cells, which corresponded to glial cells (Fig. 2). In group A, FGF-2 immunoreactivity in the neuronal perikarya and nerve fibers was greater on the *O* side of the NA than on the *U* side on day 7 after the procedure (Fig. 2A), but not on days 1, 3, 14, and 28. In group B, a dramatic change in FGF-2 immunoreactivity after transection of the RLN was observed. Increased FGF-2 immunoreactivity on the *O* side of the NA compared with the *U* side was obvious on day 7 after the procedure (Fig. 2B). In group C, however, FGF-2 immunoreactivity in the NA was slightly stronger on the *O* side than on the *U* side on day 3 after the procedure (Fig. 2C).

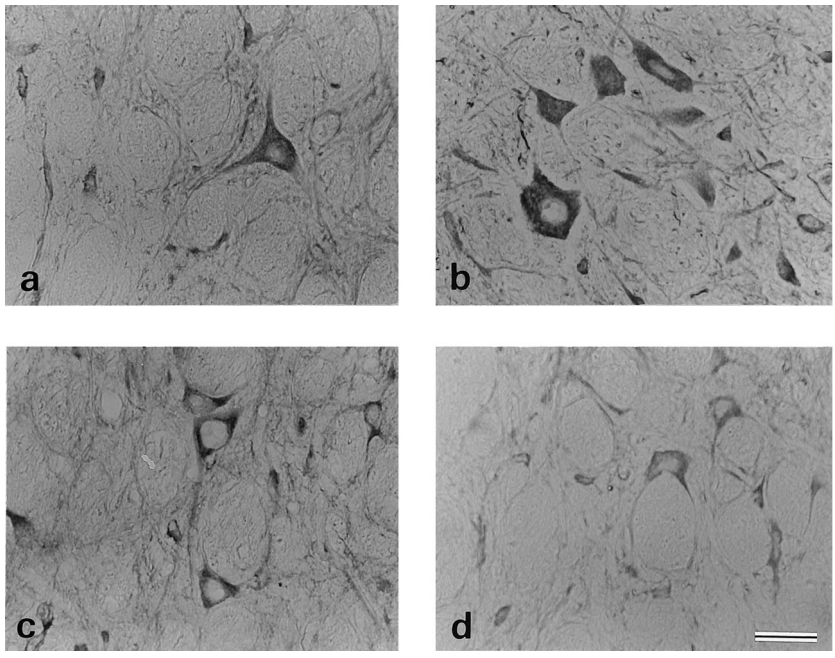


Fig. 2. Nucleus ambiguus (NA) stained with anti-fibroblast growth factor-2 (anti-FGF-2) antibody on **(A)** day 7 after nerve crush, **(B)** day 14 after nerve transection, **(C)** day 3 after nerve transection and capping, and **(D)** day 7 in unoperated controls. In group D, FGF-2 immunoreactivity was identified in the neuronal perikarya and nerve fibers of the nucleus ambiguus (NA) as brown. In the other groups, increased FGF-2 immunoreactivity in the NA was identified by enhanced staining compared with that in group D. Scale bars = 50 μm.

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Quantification of Fibroblast Growth Factor-2 Immunoreactivity

In group D, O/U was 1.03 ± 0.89, indicating the same level of FGF-2 expression in the NA bilaterally.

In group A (crush group), O/U was greater than 1.0 on days 1, 3, and 7. However, statistical significance was observed only on day 7 when O/U was 2.33 ± 0.77 (Fig. 3). Subsequently, O/U decreased to approximately 1.0.

On day 1 after the procedure in group B (transection group), O/U did not show a significant increase. Then O/U increased and peaked at 6.92 ± 0.93 on day 14. There was significantly higher FGF-2 expression on the O side on days 3, 7, and 14. O/U decreased to 1.62 ± 0.28 on day 28, with no significant difference between O and U values (Fig. 4).

In group C (transection and capping group), O/U was 1.75 ± 0.65 on day 3, indicating a significantly increased O value. The O/U was less than 1.0 on day 7 and thereafter (Fig. 5).

O/U was compared among groups A, B, and C (Fig. 6). O/U was greater in group B than in group A on days 14 and 28 and significantly greater than in group C on days 7, 14, and 28. O/U was greater in group A than in group C on days 7 and 14. There were no significant differences in O/U between groups A and B on day 7 or between groups A and C on day 28.

DISCUSSION

The intrinsic laryngeal muscles play an important role in respiration, phonation, and deglutition. They are innervated by two branches of the vagus nerve: the external branch of the superior laryngeal nerve (eSLN) and the RLN. The eSLN terminates in the cricothyroid (CT) muscle, whereas the RLN innervates the remaining intrinsic laryngeal muscles: the thyroarytenoid (TA), lateral cricoarytenoid (LCA), posterior cricoarytenoid (PCA), and arytenoid (AR) muscles. The motoneurons distributing fibers to the intrinsic laryngeal muscles through the eSLN or RLN are located in the NA, situated in the ventrolateral portion of the medulla oblongata.[15]

The motoneurons in the NA are scattered around the obex in the longitudinally elongated portion.[15] The locations of motoneurons innervating the CT, PCA, TA, LCA, and AR muscles have been demonstrated by means of several labeling techniques. Motoneurons innervating the CT muscle are located in the upper portion of the NA, whereas those innervating TA, LCA, and AR muscles are located in the lower portion of the NA. Motoneurons innervating the PCA muscle are located relatively rostrally in the NA and partly overlap those innervating the CT muscle. In this study, the brainstem was cut serially from the rostrum of the inferior olive to the caudal end of the hypoglossal nucleus. This portion includes almost all the motoneurons innervating the PCA, TA, LCA, and AR muscles and some of the motoneurons innervating the CT muscle.[15,16]

The authors quantitatively evaluated FGF-2 expression in the NA in rats with the aid of a computer-assisted image analysis system. This method provides greater objectivity, reliability, and accuracy than the subjective estimates of experts viewing stained sections.[17,18] A disadvantage of the present method of quantification is its inability to demonstrate the number of motoneurons with FGF-2 immunoreactivity or the absolute quantity of FGF-2 expression in the NA. In addition, background glial cells and interneurons were stained. Therefore, the value O/U was used as an indicator of FGF-2 expression because FGF-2 expression on the U side should not be influenced by the procedure.

This study revealed that FGF-2 immunoreactivity increased in the NA on the O side following crush and transection of the RLN (groups A and B). Nerve crush (group A) caused a vocal fold palsy, which recovered by day 28 after the crush. Based on this observation, the

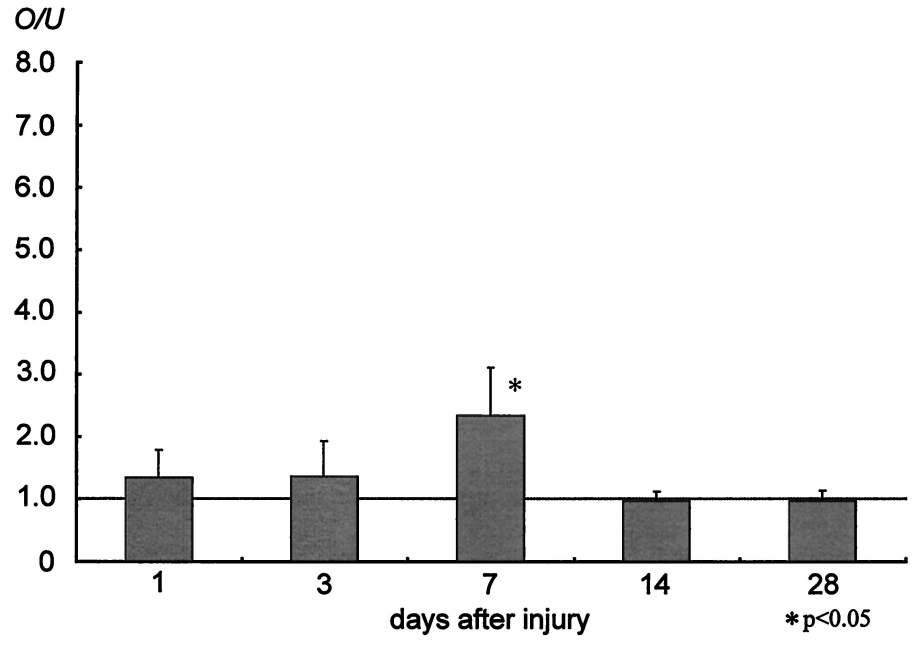


Fig. 3. FGF-2 expression after nerve crush. The number of pixels that were positive for FGF-2 immunoreactivity on the operated (*O*) and unoperated (*U*) sides were counted automatically. The FGF-2 expression in the NA induced by RLN injury was quantified as the ratio *O/U*. Height of each shaded box represents an average value of *O/U*, and the vertical bars indicate the SD (n = 6 for each observation period). *O/U* was increased on days 1, 3, and 7. However, only the increase on day 7 was statistically significant ($P < .05$, Student *t* test).

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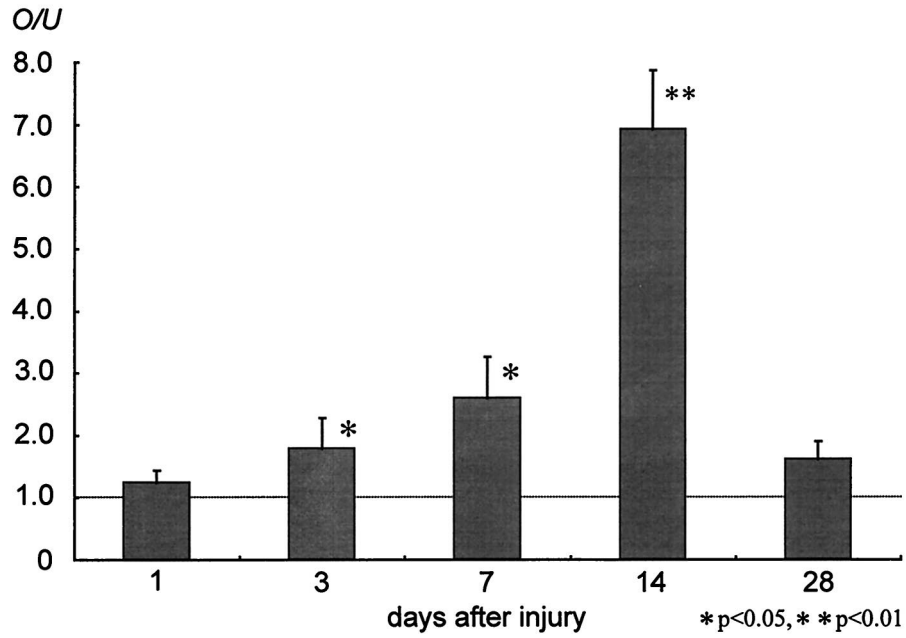


Fig. 4. FGF-2 expression after nerve transection. There was no significant difference between *O* and *U* on days 1 and 28. *O/U* indicated significantly higher FGF-2 expression on the *O* side on days 3, 7 ($P < .05$, Student *t* test), and 14 ($P < .01$, Student *t* test) after the procedure.

nerve damage attributable to the crush was considered to be Sunderland grade II damage.[14] Previous in vitro studies have shown that FGF-2 promotes neuronal survival and neurite extension in cultures of neurons from various regions of the central[19–21] and peripheral[22] nervous systems. In vivo studies using experimental paradigms of central and peripheral nerve injuries revealed that exogenously applied FGF-2 facilitated axonal regeneration. Neuronal death was caused when axonal regeneration did not occur.[9,23] It usually takes approximately 2 weeks after nerve transection until neuronal death occurs in adult rats.[3,4,24] High levels of FGF-2 expression in the NA after RLN injuries in groups A and B demonstrated elevated production of endogenous FGF-2. The elevated production of FGF-2 might contribute to preventing lesion-induced neuronal death as described previously,[9,23,25,26] although we did not assess the number of motoneurons.

The FGF-2 expression was significantly greater in group B than in group A. Using adult rats, Huber et al.[11] detected FGF-2 messenger RNA (mRNA) and reported a higher level of FGF-2 mRNA in the hypoglossal nucleus following hypoglossal nerve injury in the nerve transection group than in the crush group. He inferred increased FGF-2 expression based on the results obtained from FGF-2 mRNA measurements. These findings suggest that increased FGF-2 expression in the nucleus should be

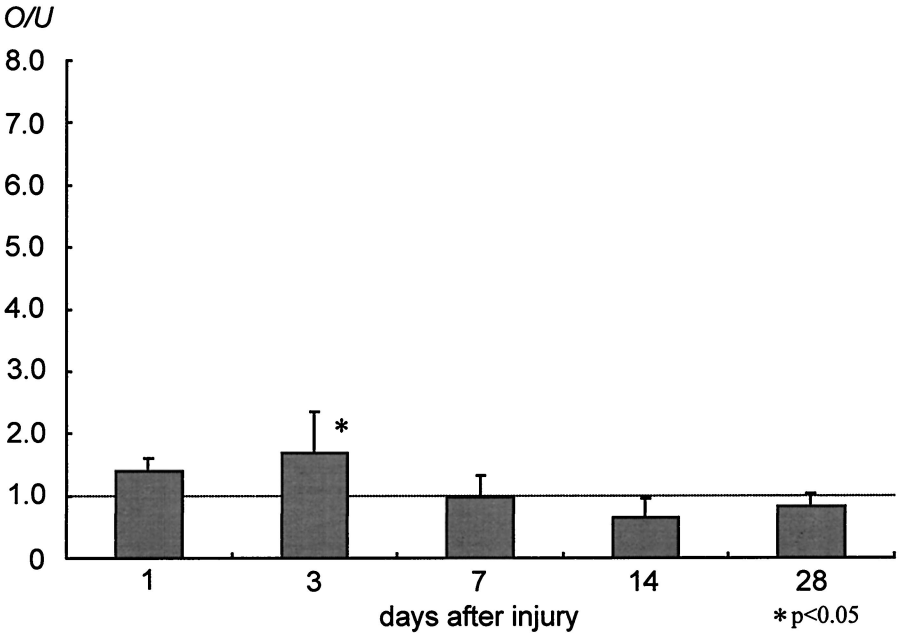


Fig. 5. FGF-2 expression after nerve transection and capping. *O/U* was 1.75 ± 0.65 on day 3, indicating a significantly increased *O* ($P < .05$, Student *t* test). *O/U* was less than 1.0 on day 7 and thereafter.

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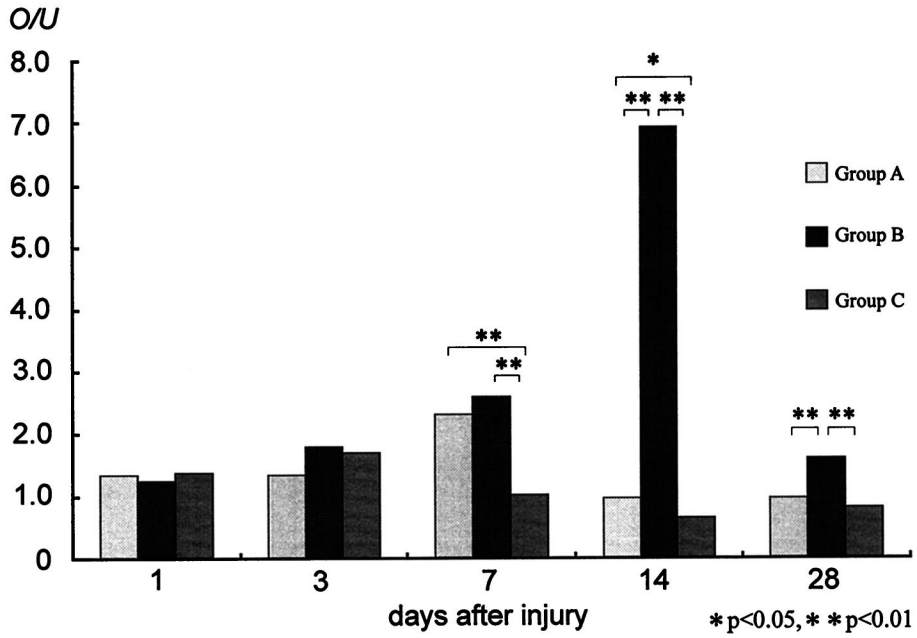


Fig. 6. Comparison of *O/U* values among groups A, B, and C. *O/U* was significantly greater in group B than in group A on days 14 and 28 ($P < .01$, unpaired Student *t* test) and than in group C on days 7, 14, and 28 ($P < .01$, unpaired Student *t* test). *O/U* was larger in group A than in group C on days 7 and 14 ($P < .01$ and $P < .05$, respectively; unpaired Student *t* test).

closely associated with the severity of nerve injury. Previous studies reported that nerve crush rarely induced motoneuron death in adult animals,[27–29] whereas nerve transection induced death of 7% to 23% of motoneurons in the sciatic and facial nerves.[3,4] Severe damage of the nerve requires a longer time for reinnervation, resulting in a higher risk of neuronal death. Therefore, higher FGF-2 expression in the nucleus is necessary for neuronal survival.

In group C, silicone caps fixed on the nerve stumps prevented reinnervation of the transected nerve. In this group, the increase in FGF-2 expression on the *O* side was minimal compared with that on the *U* side throughout the study period, except on day 3 after the procedure. The peak of FGF-2 expression was seen on days 7, 14, and 3, respectively, in groups A, B, and C. The maximal level of FGF-2 expression was significantly lower in group C than in groups A and B, although the animals in groups B and C both underwent nerve transection. These differences may be explained in the following way. Axonal sprouting begins approximately 3 hours after nerve injury, and the regenerating axons gradually grow for 5 days, irrespective of the presence or absence of their target.[30] Increased FGF-2 expression in the motoneuron is considered indispensable for motoneuron regeneration and survival. It increases exclusively in the early stage of regeneration, as shown in Figures 3 and 4. After reinnervation or neuronal death occurs, FGF-2 expression decreases in the motoneuron. According to Fujimoto et al.,[31] the regenerating axon and Schwann cells promote the local production of FGF-2. FGF-2 of peripheral origin may be transported to the nucleus by axonal flow. Because the silicone caps blocked axonal flow from the stump in group C, FGF-2 expression may have been suppressed in the NA.

CONCLUSION

We examined FGF-2 expression in the NA after three different nerve injury treatments in adult rats: nerve crush and transection of the RLN with and without covering each nerve stump with a silicone cap. FGF-2 immunoreactivity increased in the NA on the *O* side after crush (group A) and transection (group B) of the RLN. The expression of FGF-2 was significantly greater in group B than in group A. Severe damage of the nerve requires a longer time for reinnervation, resulting in a higher risk of neuronal death. FGF-2 expression in group C increased minimally throughout the experiment, except on day 3 after the procedure. By day 7, regeneration of the axon was blocked, which may have suppressed FGF-2 expression in the NA. The absence of axonal transport to motoneurons are considered to be a major reason for a relatively low level of FGF-2 expression in the NA, although a decrease in the number of motoneurons might have resulted in decreased FGF-2 production in the NA. Further study is necessary to achieve our ultimate goal of determining the role of FGF-2 in preventing neuronal death and facilitating axonal regeneration after varying degrees of RLN injury.

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